

WORKING_PLAN

HelixTranslate — Working Plan: Unfinished Items & Known Issues

Revision: 1 **Last modified:** 2026-06-14T18:10:00Z **Authority:** operator mandate 2026-06-14 ("what is left unfinished and what are (known) issues ... nothing left unfinished with zero issues when we finally finish") **Scope:** the complete, honest (§11.4.6) inventory of everything not-done + every known issue, structured as a subagent-driven execution plan. Each item carries: WHAT, WHY-OPEN/BLOCKER, EVIDENCE, SUBAGENT-TASK, ACCEPTANCE (rock-solid proof per §11.4.123, no bluff).

Baseline at authoring: HEAD 5b0dd17; go build ./... clean; full go test ./... -p 1 GREEN (57 ok / 0 FAIL at HEAD daa2f70; HEAD 5b0dd17 adds only llm.go fixes — pkg/distributed/TestSSHPool_cleanup is a confirmed full-sweep-load flake, passes -count=3 isolated, §11.4.7). 27 genuine mutation-proven bug fixes landed this session across 10 parallel subagent waves; the main-module product bug-hunt is **saturated** per the §11.4.118 completeness-critic audit.

Status legend: OPEN · OPERATOR-BLOCKED (needs credentials/decision) · DESIGN (needs architecture decision) · AUTONOMOUS (a subagent can close it now) · DONE.

P0 — Release blockers (must close before any release tag)

P0.1 — Version string inconsistency [AUTONOMOUS]

- **WHAT:** Authoritative version is ambiguous. VERSION=2.3.0; cmd/grpc-server & cmd/unified-translator appVersion=3.0.0; cmd/translator appVersion=2.1.0; CLAUDE.md says treat VERSION (2.3.0) as authoritative, Makefile references 3.0.0.
- **EVIDENCE:** cat VERSION=2.3.0; grep appVersion cmd/*/main.go → 3.0.0/3.0.0/2.1.0.
- **SUBAGENT-TASK:** Reconcile every binary's appVersion to the single authoritative VERSION value (read VERSION at build time or a shared pkg/version constant); remove the hardcoded divergent literals. RED test asserting each appVersion == VERSION.

- **ACCEPTANCE:** all appVersion literals equal VERSION; mutation (revert one) → test FAILs. **NOTE:** the authoritative number (2.3.0 vs 3.0.0) is an **operator decision** — see P1.0; the subagent wires the single-source-of-truth, operator picks the value.

P0.2 — Full §11.4.40 release retest not yet run [OPERATOR-BLOCKED on devices/topology]

- **WHAT:** Release requires the complete §11.4.40 7-step retest (pre-build sweep, post-build sweep, on-device cycle, meta-test mutation sweep, Challenge bank sweep, Issues/Fixed audit, CONTINUATION sync) on a clean baseline. We have run repeated full `go test ./... -p 1` sweeps (green) but not the full 7-step ritual.
- **SUBAGENT-TASK:** none yet — gated on P4 (gates), P6 (Challenges/HelixQA), and the §11.4.151 release-prefix tagging. Sequenced AFTER P4/P6.
- **ACCEPTANCE:** all 7 steps GREEN with captured evidence + operator confirmation + §11.4.151 `helix_translate-<version>` tag pushed to all upstreams (no force §11.4.113).

P0.3 — No §11.4.151 prefixed release tag yet [OPERATOR-BLOCKED]

- **WHAT:** No release tag exists. Per §11.4.151 the tag must be `helix_translate-<version>` (prefix from `HELIX_RELEASE_PREFIX` env or lowercased root dir name) on main repo + every owned submodule.
 - **ACCEPTANCE:** prefixed tag created post-P0.2, fanned to all upstreams.
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P1 — Operator / credential-gated (cannot be closed autonomously; need the operator)

P1.0 — Decide the authoritative version number [OPERATOR-BLOCKED]

- **WHAT:** Is the next version 2.3.x or 3.0.0? CLAUDE.md says VERSION (2.3.0) wins; Makefile/binaries say 3.0.0. Operator must pick.

P1.1 — Provider credentials absent/invalid [OPERATOR-BLOCKED]

- **OPENAI_API_KEY** absent → OpenAI provider unverified; allowlist stale (`gpt-3.5-turbo/gpt-4/gpt-4-turbo/gpt-4o` only — missing `gpt-4o-mini/4.1/o-series`).
- **ANTHROPIC_API_KEY** absent → Anthropic unverified; allowlist very stale (only `claude-3 2024` models; missing `3.5/3.7/4.x`).
- **GEMINI_API_KEY** invalid ("API Key not found", re-confirmed via live / models) → gemini non-functional.
- **ZHIPU** account out of balance (error 1113 "余额不足") → allowlist stale (live / models = `glm-4.5/4.5-air/4.6/4.7/5/5-turbo/5.1`; our allowlist has none of them) but **cannot verify translation or response-shape** → fix blocked.

- **UNBLOCK:** operator adds/refreshes the keys / recharges Zhipu. Then the **deepseek pattern** applies (proven this session): live /models → verify-translate + verify string-content shape → additive allowlist update + RED-proven gate guard.
- **EVIDENCE:** live /models probes captured this session; deepseek fix 0fd1a34 is the template.

P1.2 — ~30 other provider allowlists not audited [OPERATOR-BLOCKED on keys]

- **WHAT:** qwen, groq, cohere, mistral (live shows magistral-* reasoning family missing — but magistral-medium returns structured-list content our client can't parse, see P2.4), xai, replicate, cerebras, cloudflare, siliconflow, hyperbolic, togetherai, sambanova, kimi, novita, nlpcloud, upstage, sarvam, modal, publicai, nia, vulavula — allowlists not verified against live current models.
 - **UNBLOCK:** funded keys per provider → apply the deepseek pattern per provider (verify-translate + string-content shape before adding).
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P2 — Design decisions (need architecture/operator input; not blind autonomous fixes)

P2.1 — Inert CLI flags [DESIGN]

- **WHAT:** cmd/unified-translator parses -chunk-size, -workers, -concurrency, -verify but **never consumes** them (real chunking is automatic + correct via translateWithRetry/splitText). startMonitoringServer is a print-only stub (the -monitoring flag does not start a real monitor in the unified CLI).
- **DECISION NEEDED:** wire each flag to real behavior (define semantics) OR remove it (§11.4.122 — removing a user-facing flag needs operator confirmation). Removing/altering is not a blind fix.
- **SUBAGENT-TASK (after decision):** wire or remove per operator choice, with RED→GREEN tests.

P2.2 — Inert config fields [DESIGN]

- **WHAT:** DOCXConfig.MinTextLength + DOCXConfig.IgnoreStyles and PDFConfig.MinTextLength are declared/documented but never consumed.
- **DECISION:** wire (filter short paragraphs / honor ignore-styles) or drop. Wiring MinTextLength changes output → needs care + tests.

P2.3 — Verifier MinScoreThreshold scale inconsistency [DESIGN]

- **WHAT:** the handler (pkg/api/verifier_handlers.go) compares MinScoreThreshold against **raw 0-100** OverallScore; the adapter (internal/services/llmsverifier_score_adapter.go) compares it against the **normalized 0-10** score. The two contracts contradict; GetPreferences/GetProviderScore currently have **no production caller**.

- **DECISION NEEDED:** declare the canonical scale (0-100 or 0-10) before either path is wired. Fixing either side blindly breaks the other's passing test (§11.4.120).

P2.4 — Reasoning-model structured-content support [DESIGN]

- **WHAT:** OpenAI-compatible clients assume content is a STRING. Some reasoning models return content as a structured LIST (verified: Mistral magistral-medium-latest; likely glm-5 / deepseek-reasoner class). Our clients silently drop / can't parse it.
- **DECISION:** add structured-content handling to the OpenAI-compatible client layer → would unlock magistral/glm-5/reasoner models. Non-trivial; design + tests required.

P2.5 — Markdown not a first-class CLI input [DESIGN]

- **WHAT:** .md input is detected as TXT and translated as plain text (works, but markdown structure not preserved). First-class markdown input is an enhancement.

P2.6 — cmd/translator intermediate-markdown download-dir inconsistency [OPEN, needs live SSH]

- **WHAT:** intermediate .md downloads to Dir(InputFile) in one path vs Dir(OutputFile) in another; manifests only under live SSH with -o in a different dir. Not unit-testable without real SSH infra.
- **SUBAGENT-TASK:** reproduce via the §11.4.76 Containers submodule (boot an SSH worker container) → RED → fix → GREEN. Gated on containerized SSH test infra.

P3 — Dead / unwired code (§11.4.124 investigate-before-remove)

P3.1 — pkg/hash package is dead [AUTONOMOUS – investigate]

- **WHAT:** pkg/hash (393 LOC) has **ZERO importers** across the module (confirmed). Per §11.4.124 it must NOT be removed without git-history investigation establishing how/when it became dead + whether a hidden reference exists.
- **SUBAGENT-TASK:** git-history investigation (`git log --follow, -S pickaxe, blame`) → capture as FACT where it was wired + when it died → either (a) restore a mistakenly-deleted call-site / wire it in properly + add tests, or (b) if proven genuinely unneeded, remove it in its own descriptive commit citing the git evidence (+ operator confirm per §11.4.122 since it's a shipped package).
- **ACCEPTANCE:** captured git-history evidence + decision; if removed, separate commit citing evidence.

P4 — Governance / constitution-compliance gaps (large; the project under-implements many mandates)

P4.1 — Workable-items tracker constellation missing [AUTONOMOUS scaffold + ongoing]

- **WHAT:** The constitution mandates docs/Issues.md, docs/Issues_Summary.md, docs/Fixed.md, docs/Fixed_Summary.md, docs/CONTINUATION.md (✓ exists), per-item ATM-NNN tickets (§11.4.54), the SQLite docs/workable_items.db single-source-of-truth (§11.4.93/.95), procedure docs docs/procedures/issues/*.md (§11.4.63), Reopens/Status docs. **CONFIRMED ABSENT:** Issues/Fixed/summaries, procedures/, workable_items.db, ATM tickets. The project tracks work in CONTINUATION.md only.
- **SUBAGENT-TASK:** scaffold the tracker constellation: migrate this session's 27 fixes into Fixed.md + Fixed_Summary.md with ATM-NNN IDs + Status/Type columns; create empty Issues.md/Issues_Summary.md; stand up the §11.4.93 Go cmd/workable-items tool + DB (or document a deliberate §11.4.6 deviation). Large — split across sub-tasks.
- **ACCEPTANCE:** the doc-set exists, in sync (.md + .html + .pdf per §11.4.65), gates green.

P4.2 — Pre-build CM-* gate suite not implemented [AUTONOMOUS, large]

- **WHAT:** The constitution references dozens of CM-* pre-build gates + paired §1.1 mutations. This project has only scripts/testing/meta_test_constitution_inheritance.sh; the CM-* gate suite + pre_build_verification.sh are NOT implemented.
- **SUBAGENT-TASK:** implement the highest-value gates first (anti-bluff smoke, doc-sync, regression-guard-registered, no-fakes-beyond-unit, gitignore-precommit-audit) each with a paired mutation, wired into a scripts/pre_build_verification.sh.
- **ACCEPTANCE:** each gate present + its paired mutation FAILs it; runnable as a suite.

P4.3 — §11.4.65 universal markdown export audit [AUTONOMOUS]

- **WHAT:** all tracked non-source .md must have synced .html+.pdf siblings. The commit wrapper auto-syncs, but a full audit (incl. this new WORKING_PLAN.md + all docs/) is needed.
 - **SUBAGENT-TASK:** run/verify sync_all_markdown_exports-equivalent; confirm every docs/*.md has fresh .html/.pdf.
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P5 — Owned submodules (§11.4.28 equal-codebase) [OPERATOR-COORDINATION]

P5.1 — Submodule bug-hunt + brittle-test fixes [BLOCKED: another session active]

- **WHAT:** Owned submodules present: challenges, containers, helix_qa, doc_processor, llm_orchestrator, llm_provider, vision_engine, llms_verifier, docs_chain. They are §11.4.28 equal-codebase and have the brittle "connection refused" env-coupled test class (15+ sites in challenges/containers/helix_qa) — the same class fixed in the main module this session.
 - **BLOCKER:** evidence shows **another session is actively working helix_qa** (observed `github.com/HelixDevelopment/... go test -race` running). Per §11.4.119 single-owner, do NOT collide.
 - **SUBAGENT-TASK (after operator confirms ownership/coordination):** per-submodule bug-hunt waves (separate `go.mod`, separate upstreams, their own commit flow) mirroring the main-module campaign.
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P6 — Full test-type coverage (§11.4.25 / §11.4.27) [AUTONOMOUS + infra-gated]

P6.1 — Per-feature test-type matrix + HelixQA + Challenges [OPEN, large]

- **WHAT:** §11.4.27 mandates 100% coverage with every test type (unit ✓, integration ✓ -partial, e2e ✓ -partial, security ✓ -partial, stress/chaos ✓ -partial via §11.4.85, plus ddos/scaling/perf/benchmark/ui/ux, Challenges bank, full HelixQA autonomous sessions). This session added unit/integration + some stress/chaos + real E2E proofs; the full matrix per feature is unfilled.
- **SUBAGENT-TASK:** build the §11.4.25 coverage ledger (feature × platform × test-type × status); fill the highest-value gaps (perf/benchmark for the translation pipeline, chaos for distributed/storage, Challenges entries per shipped feature in `tools/helixqa/banks/`).
- **ACCEPTANCE:** coverage ledger published; each shipped feature has ≥ the mandated test types with captured evidence under `docs/qa/<run-id>/` (§11.4.83).

P6.2 — docs/qa/ evidence per shipped feature [AUTONOMOUS]

- **WHAT:** §11.4.83 requires a recorded e2e transcript per shipped feature. We have E2E proofs for PDF input, output-format matrix, deepseek-v4. The 27 fixes' user-visible features need per-feature evidence dirs.
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Confirmed GREEN / DONE this session (for completeness)

- 27 genuine mutation-proven bug fixes (waves 1-10) — see CONTINUATION.md Rev 36-41 for the itemized list with commit hashes.
 - Format matrix complete: 6 inputs (FB2/EPUB/TXT/HTML/DOCX/PDF) × 5 outputs (EPUB/FB2/HTML/TXT/MD), all real-translation proven.
 - PDF + DOCX input revived from license-gated-dead (MIT ledongthuc/pdf + stdlib OOXML).
 - DeepSeek allowlist current (v4 models, live-/models-proven).
 - All sweeps GREEN at quiescent checkpoints (57 ok / 0 FAIL); build+vet clean; all FF-pushed to both upstreams (no force §11.4.113).
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Execution order (recommended)

1. **Now (autonomous):** P0.1 version single-source wiring (pending P1.0 value) · P3.1 pkg/hash investigation · P4.3 export audit · P6.2 evidence backfill.
2. **After operator input:** P1.0 version number · P1.1/P1.2 keys → allowlist audits · P2.x design decisions.
3. **Larger autonomous tracks:** P4.1 tracker constellation · P4.2 CM-gate suite · P6.1 test-type matrix.
4. **Coordinated:** P5.1 submodules (confirm no collision).
5. **Final:** P0.2 full §11.4.40 retest → P0.3 §11.4.151 prefixed release tag.

Nothing here is silently dropped. Every item is tracked to a subagent task with rock-solid-proof acceptance (§11.4.123) and no-bluff verification (§11.4).